

Dhafer Malouche, Ph.D.

Professor of Statistics

Qatar University
✉ dhafer.malouche@qu.edu.qa
🌐 dhafermalouche.net
👤 [malouche](#)
Google Scholar Profile

Employment History

- August 2022 – Present **Professor of Statistics, Qatar University, Doha, Qatar**
Department of Mathematics, Statistics, and Physics, College of Arts and Sciences.
Teaching (2022-2025): Introduction to Statistics, Multivariate Analysis, Stochastic Processes, Actuarial Statistics I&II, Statistical Computation and Simulation, Advanced Biostatistics (For Ph.D. Candidates), Bayesian Statistical, Experimental Design, Critical Thinking (Honors Program), Expertise, Experience and Exchange: tips for research methods (Honors Program).
- July 2021 – July 2022 **Professor of Statistics, The American University in Cairo, Cairo, Egypt**
Department of Mathematics and Actuarial Science, School of Sciences and Engineering.
Teaching: Statistical Inference, Introduction to Statistics, Data Science with Python
- 2003 – 2021 **Professor / Associate Professor / Assistant Professor, University of Carthage, Tunis, Tunisia**
Ecole Supérieure de la Statistique et de l'Analyse de l'Information
Positions: Professor (from 2018), Associate Professor (2011-2018), Assistant Professor (2003-2011).
Teaching: Data Analysis: Principal Component Analysis, Correspondence Analysis, and Multiple Correspondence Analysis, Theory and Practice with R, Mathematical Statistics: Statistical Inference, Hypothesis Testing, Regression Analysis, Theory and Practice with R, Data Mining and Practice with R, Bayesian Statistics with OpenBugs/Winbugs/RStan/Jags, Time Series: ARMA and SARIMA Processes, Theory and Practice with R/Python, Big Data: Large Data with R/Python, SQL, Spark, H2O, Advanced R/Python: Data Management, Data Visualization, Dashboards, Shiny Apps, Heroku Apps, Bokeh.
Research: Supervising 5 Ph.D.: Detection and classification of swallowing sound, Sensory Analysis, Genetics, Zoonotic cutaneous leishmaniasis incidence, Cardiovascular risk factor. Supervising Master and Engineering Thesis on several topics related to applied statistics and data science.
Administration: Director of the Department of Statistics (2004-2007)
- 2014 and 2019 **Visiting Associate Professor / Associate Research Scholar, Yale University, New Haven, USA**
Whitney and Betty MacMillan Center for International and Area Studies and the Department of Statistics and Data Science.
Positions: Visiting Associate Professor (2014), Consulting on several projects with Yale Scholars (2015-2018), Associate Research Scholar (2019).
Missions: Democratic Transition in Tunisia, Governance and Local Development: Implementing two face-to-face surveys in Tunisia, G-econ team on building local GDP data, Teaching Time Series with R/Python Course.
- 2016 – 2017 **Visiting Fulbright Scholar, University of Michigan, Ann Arbor, USA**
Center for Political Studies of the Institute for Social Research.
Role: Research: Working on Data Quality, Survey Methodology, Interviewer Effect. Teaching: 4 Lectures on Applied graphical Models, 1 Lecture on Data visualization with R, 1 Lecture on Sensory Analysis.
- 2011 **Visiting Fulbright Scholar, Stanford University, Palo Alto, USA**
Department of Statistics.
Role: Research on Graphical Models, Faithfulness assumption, and Covariance graphs.

- 2002 – 2003 **Visiting Assistant Professor**, *York University*, Toronto, Canada
 Department of Statistics.
Teaching: Applied Regression Models with SAS, Introduction to the theory of probability, Introduction to Statistics with Minitab.
Research: Monte Carlo Methods and Bayesian Estimation of the Graphical Models.
- 1998 – 2002 **Assistant Professor**, *University of Sousse*, Sousse, Tunisia
 Institut Préparatoire aux Ecoles d'Ingénieurs.
Teaching: Analysis, Calculus, Algebra.
Research: Natural Exponential Families, Pick functions, Markov Chains.
Administration: Director of the Department of Mathematics

Education

- September 2009 **Habilitation (Tenure), Statistics**, *Université de Tunis El Manar*, Tunisia
 Ecole National d'Ingénieurs de Tunis.
 Thesis title: *Problèmes autour de la probabilité et de la statistique: Méthodes et Applications*.
 Dissertation: <https://malouche.github.io/myCV/reports.html>
- October 1997 **Doctorate (Ph.D.), Statistics**, *Paul Sabatier University*, Toulouse, France
 Thesis title: *Classification des familles exponentielles associées à des fonctions Pick*.
 Dissertation: <https://malouche.github.io/myCV/reports.html>
- 1993 – 1994 **Master's Degree (D.E.A)**, *Paul Sabatier University*, Toulouse, France
 Applied Mathematics, Statistics
- 1989 – 1993 **Bachelor (Maîtrise)**, *Ecole Normale Supérieure de Bizerte*, Tunisia

Supervised Ph.D.

- 2019-06-28 **Safa AOUINTI**
 Statistical treatment of NGS results from IMGT/HighV-QUEST of antigen receptors (immunoglobulins and T receptors): methodology and visualization
- 2019-06-22 **Khouloud TALMOUDI JABBARI**
 Spatio-Temporal Modeling of Zoonotic Cutaneous Leishmaniasis (ZCL) in Central Tunisia
- 2019-05-21 **Hajer KHLAIFI**
 Preliminary study on the detection and classification of swallowing sounds
- 2019-02-14 **Oifa SAIDI**
 Modeling the Tunisian Burden of Cardiovascular Diseases and Diabetes
- 2018-07-02 **Ibtihel REBHI**
 Statistical methods for sensory and consumer data mapping

Skills

- Languages Strong reading, writing and speaking competencies for English, French, and Arabic.
- Coding Python, R, tableau, spark, h2o, Shiny, $\text{\LaTeX}$, ...
- Misc. Quantitative research, Project management, Qualitative research, Data Mining, Machine Learning, Big Data, Academic research, Teaching, training, consultation, $\text{\LaTeX}$ typesetting and publishing.

Grants

- 2024-2025 **Autoimmune disorders profile pre and post-COVID era**, *QF-QBB-RES-ACC-00277*
 Research on the prevalence and management of autoimmune diseases in Qatar using advanced data analysis techniques to develop prognostic models and inform public health policies, based on Qatar Biobank data from 2015-2023.

- 2024-2025 **Investigating Gender Disparities in Quality of Life Across the Qatari Population, QF-QBB-RES-ACC-00281**
Research project analyzing health-related quality of life (HRQOL) disparities in Qatar, focusing on gender differences using data from the Qatar Bio Bank. The project utilizes advanced machine learning to inform public health policies and reduce health inequalities.
- Fall 2025 **Proteomic Signatures of Fatigue and Mental-Health Symptoms in the Qatar Biobank Cohort, QF-QBB-RES-ACC-00178**
Research project investigating molecular mechanisms underlying chronic fatigue in Qatar Biobank participants. Using proteomic profiling to identify inflammatory and metabolic signatures that differentiate fatigued from non-fatigued individuals, with integration into Bayesian networks to test pain-fatigue-mental health pathways and explore sex-specific profiles for targeted interventions.
- Fall 2024 **Detecting the Undetectable: A Study on AI Tool Efficacy in Academic Writing, Student Grant 2024 Cycle 2, 130**
Research project evaluating the efficacy of AI detection tools in identifying AI-generated text in academic settings, aiming to enhance academic integrity policies at Qatar University.
- Spring 2025 **Comparative Analysis of Research Publication Trends, Student Grant 2025 Cycle 1, 293**
A Decade-Long Study of Leading MENA and Global Universities (2010–2020). This study aims to analyze research output trends, citation impact, and collaboration networks among major universities in the MENA region and globally.

Papers

Research Interests

Bayesian networks and graphical models, Biostatistics and health outcomes research, Qatar Biobank data analysis, Path analysis and causal inference, Meta-analysis and systematic reviews, Chronic pain and mental health modeling, Educational statistics, Statistical computing and data visualization, Consumer preferences and sensory data analysis.

Research Papers

1. Naija, A., Mohamed, O., Malouche, D., & Abiib, S. (2026). Global phthalate occurrence and risk assessment in drinking water: a comprehensive Bayesian and probabilistic approach. *Science of the Total Environment*. [Accepted 3 September 2026, in press; manuscript ID STOTEN-D-26-00544R2].
2. Muhammad, M. I., Al-Khinji, A., Al-Ansari, M., Al-Marri, N., Alshaheen, S., & Malouche, D. (2026). Allele-stratified activity of RAS inhibitors in pancreatic ductal adenocarcinoma: a systematic review with model-validation-stratified synthesis of preclinical evidence and exploratory meta-analysis of early-phase trials. *Cancers*. [Accepted, in press; manuscript ID cancers-4500792].
3. Al-Khinji, A., Malouche, D., Al-Hor, A., Haj Bakri, A., & Al-Kuwari, M. G. (2026). Differential laboratory monitoring in autoimmune disease: a matched case-control study from Qatar primary care. *Journal of Translational Medicine*. [Accepted, in press; manuscript ID JTRM-D-26-01894].
4. Mzoughi, C., & Malouche, D. (2026). Legislating the future: statistical foresight as a foundation for anticipatory law. *Proceedings of the 12th Annual International Academic Conference, Kuwait International Law School (KILAW)*. [Accepted, in press].
5. Mizza, D., ElKhodary, K., Malouche, D., & Al-Mousa, M. (2026). Multi-modal virtual reality in engineering education: a design case study of fostering situational interest through project-based learning. *International Journal of Engineering Pedagogy (IJEP)*, 16(5), 4–27. doi:10.3991/ijep.v16i5.61389
6. Al-Sharman, A., Khalil, H., Malouche, D., Kanaan, S., Kim, M., Saad, N., & Abdelrazeq, M. (2026). Directional symptom dependencies in multiple sclerosis and Parkinson's disease: A comparative Bayesian network analysis. *Journal of Multidisciplinary Healthcare*. doi:10.2147/JMDH.S607760
7. Al-Khinji, A., Malouche, D., Al-Thani, N., Mustafa, A., Abdulmajeed, J., & Al-Kuwari, M. G. (2026). Hematological abnormalities in clinically diagnosed non-alcoholic steatohepatitis: prevalence, clinical correlates, and fibrosis risk in a case-control study from Qatar. *Frontiers in Medicine*, 13, 1773499. [Q1]. doi:10.3389/fmed.2026.1773499
8. Mizza, D., Reese, M., & Malouche, D. (2025). Flipped classroom evaluation and blended learning potential: a

- case study of engagement and inclusion in quantitative education. *Smart Learning Environments*, 12, 56. [Q1]. doi:10.1186/s40561-025-00412-2
9. Al Kuwari, H., Al Khinji, A., Al-Hamed, D., Al Hor, A., Malouche, D., & Iqbal, J. (2025). NOTCH1 mutation status as a prognostic biomarker in T-cell acute lymphoblastic leukemia: a systematic review and meta-analysis. *BMC Cancer*, 25(1), 1820. [Q2]. doi:10.1186/s12885-025-15233-2
 10. Al-Khinji, A., Al-Korbi, N., Al-Kuwari, S., Al-Hor, A., & Malouche, D. (2025b). Immune checkpoint inhibitors in pancreatic adenocarcinoma: A systematic review and meta analysis of clinical outcomes. *Frontiers in Oncology*, 15, 1569884. [Q2]. doi:10.3389/fonc.2025.1569884
 11. Al-Khinji, A. A. M. A., & Malouche, D. (2025c). Modeling chronic pain interconnections using bayesian networks: Insights from the qatar biobank study. *Frontiers in Pain Research*, 6. [Q1]. doi:10.3389/fpain.2025.1573465
 12. Gad, A., Malouche, D., Chhabra, M., Hoang, D., Suk, D., Ron, N., ... Elmakaty, I. (2024). Impact of birth weight to placental weight ratio and other perinatal risk factors on left ventricular dimensions in newborns: A prospective cohort analysis. *Journal of Perinatal Medicine*. [Q2]. doi:10.1515/jpm-2023-0384
 13. Khalil, H., Malouche, D., Al Marri, D., Shoukry, M., Shehabi, M., Yahia, R., ... El-Salim, K. (2024). 10. the comparative ability of five mobility and balance outcome measures in predicting the risk of falls in people with multiple sclerosis. *Multiple Sclerosis and Related Disorders*, 92, 105971. [Q2]. doi:10.1016/j.msard.2024.105971
 14. Khalil, H., Malouche, D., Kanaan, S., Al-Sharman, A., & El-Salim, K. (2024). 138. factors affecting the quality of sleep and physical activity participation of multiple sclerosis (ms) patients: A path analysis. *Multiple Sclerosis and Related Disorders*, 92, 106099. [Q2]. doi:10.1016/j.msard.2024.106099
 15. Losada-Echeberría, M., Naranjo, G., Malouche, D., Taamalli, A., Barraón-Catalán, E., & Micol, V. (2023). Influence of drying temperature and harvesting season on phenolic content and antioxidant and antiproliferative activities of olive (*olea europaea*) leaf extracts. *International Journal of Molecular Sciences*, 24(1). [Q1]. doi:10.3390/ijms24010054
 16. Malouche, D. (2023). Describing conditional independence statements using undirected graphs. *Axioms*, 12(12). [Q2]. URL
 17. Rebhi, I., & Malouche, D. (2023). Sensmap r package and sensmapgui shiny web application for sensory and consumer data mapping: Variations on external preference mapping and stability assessment. *Journal of Sensory Studies*. [Q3]. doi:10.1111/joss.12809
 18. Ben-Hassine, K., Taamalli, A., Rezig, L., Yanguì, I., Benincasa, C., Malouche, D., ... Mnif, W. (2022). Effect of processing technology on chemical, sensory, and consumers' hedonic rating of seven olive oil varieties. *Food Science & Nutrition*. [Q1]. doi:10.1002/fsn3.2717
 19. Ben-Hassine, K., Yanguì, I., Mnif, W., Taamalli, A., Benincasa, C., Kamoun, N., & Malouche, D. (2022). Chemometric analysis and physicochemical composition of foreign and tunisian olive oil: Consumer preferences. *Journal of Food Quality*, vol. 2022, Article ID 3981028, 10 pages. [Q2]. doi:10.1155/2022/3981028
 20. Malouche, D. (2022). Implication of faithfulness assumption. *Sankhya B: The Indian Journal of Statistics*. [Q4]. doi:10.1007/s13571-021-00271-0
 21. Saidi, O., Malouche, D., Saksena, P., Arfaoui, L., Talmoudi, K., Hchaichi, A., ... Ben Alaya, N. (2020). Impact of contact tracing, respect of isolation and lockdown in reducing the number of cases infected with covid-19: Case study: Tunisia's response from march 22 to 04 may 2020. *International Journal of Infectious Diseases*. [Q1]. doi:10.1016/j.ijid.2021.02.010
 22. Kongbonga, G. Y. M., Hassine, K. B., Ghalila, H., Malouche, D. et al. (2019). Front-face fluorescence using uv-led coupled to usb spectrometer to discriminate between virgin olive oil from two cultivars. *Food and Nutrition Sciences*, 10(02), 119. [Q2]. URL
 23. Mekki, I., Malouche, D., Smeti, S., Hajji, H., Mahouachi, M., M, E., & Atti, N. (2019). Study of the breeding systems of sheeps in the montagnous area of north-western tunisia. *Livestock Research for Rural Development*, 31(108). [Q3]. URL
 24. Saidi, O., Hajjem, S., Zoghlami, N., Aounallah-Skhiri, H., Mansour, N. B., Hsairi, M., ... O'Flaherty, M. et al. (2019). Premature mortality attributable to smoking among tunisian men in 2009. *Tobacco induced diseases*, 17. [Q1]. doi:10.18332/tid/112666
 25. Saidi, O., O'Flaherty, M., Zoghlami, N., Malouche, D., Capewell, S., Critchley, J. A., ... Guzman Castillo, M. (2019). Comparing strategies to prevent stroke and ischemic heart disease in the tunisian population: Markov modeling approach using a comprehensive sensitivity analysis algorithm. *Computational and mathematical methods in medicine*, 2019. [Q2]. doi:10.1155/2019/2123079

26. Saidi, O., Zoghlami, N., Bennett, K. E., Mosquera, P. A., Malouche, D., Capewell, S., . . . O'Flaherty, M. (2019). Explaining income-related inequalities in cardiovascular risk factors in tunisian adults during the last decade: Comparison of sensitivity analysis of logistic regression and wagstaff decomposition analysis. *International journal for equity in health*, 18(1), 177. [Q1]. URL
27. Salem, S., Malouche, D., & Romdhane, H. B. (2019). Tunisian population quality of life: A general analysis using sf-36. *Eastern Mediterranean Health Journal*, 25(9). [Q3]. URL
28. Khlaifi, H., Istrate, D., Demongeot, J., & Malouche, D. (2018). Swallowing sound recognition at home using gmm. *IRBM*, 39(6), 407–412. [Q4]. doi:10.1016/j.irbm.2018.10.009
29. Benstead, L. J., Kao, K., Landry, P. F., Lust, E. M., & Malouche, D. (2017). Using tablet computers to implement surveys in challenging environments. *Survey Practice*, 10(2), 1–9. [Q1]. doi:10.29115/SP-2017-0009
30. Rebhi, I., & Malouche, D. (2017a). An approach for external preference mapping improvement by denoising consumer rating data. *International Journal of Advanced Computer science and Applications*, 8(12), 500–508. [Q3]. doi:10.14569/IJACSA.2017.081266
31. Rebhi, I., & Malouche, D. (2017b). Decision making about products development through consumer preferences modeling based on descriptive characteristics of products. *IEEE/ACS 14th International Conference on Computer Systems and Applications (AICCSA)*, 423–430. doi:10.1109/AICCSA.2017.202
32. Talmoudi, K., Bellali, H., Ben-Alaya, N., Saez, M., Malouche, D., & Chahed, M. K. (2017a). Comparative performance analysis for generalized additive and generalized linear modeling in epidemiology. *International Journal of Advanced Computer Science and Applications*, 8(12). [Q3]. doi:10.14569/IJACSA.2017.081255
33. Talmoudi, K., Bellali, H., Ben-Alaya, N., Saez, M., Malouche, D., & Chahed, M. K. (2017b). Modeling zoonotic cutaneous leishmaniasis incidence in central tunisia from 2009-2015: Forecasting models using climate variables as predictors. *PLoS neglected tropical diseases*, 11(8), e0005844. [Q1]. doi:10.1371/journal.pntd.0005844
34. Triki, H. Z., Laabir, M., Lafabrie, C., Malouche, D., Bancon-Montigny, C., Gonzalez, C., . . . Daly-Yahia, O. K. (2017). Do the levels of industrial pollutants influence the distribution and abundance of dinoflagellate cysts in the recently-deposited sediment of a mediterranean coastal ecosystem? *Science of the Total Environment*, 595, 380–392. [Q1]. doi:10.1016/j.scitotenv.2017.03.183
35. Aouinti, S., Giudicelli, V., Duroux, P., Malouche, D., Kossida, S., & Lefranc, M.-P. (2016). Imgt/statclonotype for pairwise evaluation and visualization of ngs ig and tr imgt clonotype (aa) diversity or expression from imgt/highv-quest. *Frontiers in immunology*, 7, 339. [Q1]. doi:10.3389/fimmu.2016.00339
36. Aouinti, S., Malouche, D., Giudicelli, V., Kossida, S., & Lefranc, M.-P. (2016). Correction: Imgt/highv-quest statistical significance of imgt clonotype (aa) diversity per gene for standardized comparisons of next generation sequencing immunoprofiles of immunoglobulins and t cell receptors. *PloS one*, 11(1), e0146702. doi:10.1371/journal.pone.0146702
37. Kerfai, N., Bejar Ghadhab, B., & Malouche, D. (2016). Performance measurement and quality costing in tunisian manufacturing companies. *The TQM Journal*, 28(4), 588–596. [Q1]. URL
38. Saidi, O., Malouche, D., O'Flaherty, M., Mansour, N. B., Skhiri, H., Romdhane, H. B., & Bezdah, L. (2016). Assessment of cardiovascular risk in tunisia: Applying the framingham risk score to national survey data. *BMJ open*, 6(11), e009195. [Q1]. doi:10.1136/bmjopen-2015-009195
39. Selmi, G., Azouz, Z. B., & Malouche, D. (2015). The volume radius function: A new descriptor for the segmentation of volumetric medical images. *2015 International Conference on Image and Vision Computing New Zealand (IVCNZ)*, 1–6. doi:10.1109/IVCNZ.2015.7761572
40. Aouinti, S., Malouche, D., Giudicelli, V., Kossida, S., & Lefranc, M.-P. (2015). Imgt/highv-quest statistical significance of imgt clonotype (aa) diversity per gene for standardized comparisons of next generation sequencing immunoprofiles of immunoglobulins and t cell receptors. *PLoS One*, 10(11), e0142353. [Q1]. URL
41. Hassine, K. B., Taamalli, A., Slama, M. B., Khoulood, T., Kiristakis, A., Benincasa, C., . . . Bornaz, S. et al. (2015). Characterization and preference mapping of autochthonous and introduced olive oil cultivars in tunisia. *European Journal of Lipid Science and Technology*, 117(1), 112–121. [Q2]. doi:10.1002/ejlt.201400049
42. Saidi, O., O'Flaherty, M., Mansour, N. B., Aissi, W., Lassoued, O., Capewell, S., . . . Romdhane, H. B. (2015). Forecasting tunisian type 2 diabetes prevalence to 2027: Validation of a simple model. *BMC public health*, 15(1), 104. [Q1]. URL
43. Ghribi, K., Sevestre, S., Guessoum, Z., Gil-Quijano, J., Malouche, D., & Youssef, A. (2014). A survey on multi-agent management approaches in the context of intelligent energy systems. *2014 International Conference on Electrical Sciences and Technologies in Maghreb (CISTEM)*, 1–8. doi:10.1109/CISTEM.2014.7077030

44. Hassine, K. B., El Riachy, M., Taamalli, A., Malouche, D., Ayadi, M., Talmoudi, K., ... Romano, E. et al. (2014). Consumer discrimination of chemlali and arbequina olive oil cultivars according to their cultivar, geographical origin, and processing system. *European journal of lipid science and technology*, 116(7), 812–824. [Q2]. doi:10.1002/ejlt.201300254
45. Karaoud, M., Bouafif, N., Malouche, D., Kouni, C., & Achour, N. (2014). La mortalité parmi les enfants âgés de moins de 15 ans en tunisie peut être liée à la température. *Revue d'Épidémiologie et de Santé Publique*, 62, S122. [Q3]. URL
46. Karaoud, M., Malouche, D. et al. (2014). Les effets de la température sur la mortalité chez les personnes âgées en tunisie. *Revue d'Épidémiologie et de Santé Publique*, 62, S219–S220. [Q3]. URL
47. Karaoud, M., Malouche, D., & Alaya, N. B. (2014). Mortalité et température journalière en tunisie: Une étude multi-région. *Revue d'Épidémiologie et de Santé Publique*, 62, S218. [Q3]. URL
48. Karaoud, M., Malouche, D., & Bouafif, N. (2014). Méthodologie de l'analyse de la relation température–mortalité en tunisie. *Revue d'Épidémiologie et de Santé Publique*, 62, S141–S142. [Q3]. URL
49. Triki, H. Z., Daly-Yahia, O. K., Malouche, D., Komiha, Y., Deidun, A., Brahim, M. et al. (2014). Distribution of resting cysts of the potentially toxic dinoflagellate alexandrium pseudogonyaulax in recently-deposited sediment within bizerte lagoon (mediterranean coast, tunisia). *Marine pollution bulletin*, 84(1-2), 172–181. [Q1]. doi:10.1016/j.marpolbul.2014.05.014
50. Aouinti, S., Mallek, H., Malouche, D., Saidi, O., Lassouedi, O., Hentati, F., & Romdhane, H. B. (2013). Graphical interaction models to extract predictive risk factors of the cost of managing stroke in tunisia. *2013 International Conference on Computer Medical Applications (ICCMA)*, 1–6. URL
51. Ben-Hassine, K., Taamalli, A., Ferchichi, S., Mlaouah, A., Benincasa, C., Romano, E., ... Hammami, M. (2013). Physicochemical and sensory characteristics of virgin olive oils in relation to cultivar, extraction system and storage conditions. *Food research international*, 54(2), 1915–1925. [Q1]. URL
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53. Hassine, K. B., Taamalli, A., Malouche, D., Kammoun, N., Lazzez, A., Benincasa, C., ... Bouaziz, M. (2012). Influence of variety, geographical site and extraction system on virgin olive oil (voo) linoleic acid composition and its impact on consumer preference. *Linoleic Acids*, 97. [Q2]. URL
54. Malouche, D., & Rajaratnam, B. (2011). Gaussian covariance faithful markov trees. *Journal of Probability and Statistics*, 2011. [Q3]. doi:10.1155/2011/152942
55. Ranjanomennahary, P., Ghalila, S. S., Malouche, D., Marchadier, A., Rachidi, M., Benhamou, C., & Chappard, C. (2011). Comparison of radiograph-based texture analysis and bone mineral density with three-dimensional microarchitecture of trabecular bone. *Medical Physics*, 38(1), 420–428. [Q1]. doi:10.1118/1.3528125
56. Arfa, I., Nouria, S., Abid, A., Alaya, N. B.-B., Zorgati, M., Malouche, D., ... Romdhane, H. B. et al. (2010). Absence d'association entre les polymorphismes du système rénine angiotensine (sra) et l'hypertension artérielle chez les diabétiques de type 2 tunisiens. *la tunisie médicale*, 88(01), 37–40. [Q4].
57. Arfa, I., Nouria, S., Abid, A., Bouafif-Ben, N. A., Zorgati, M. M., Malouche, D., ... Ben, H. R. et al. (2010). Lack of association between renin-angiotensin system (ras) polymorphisms and hypertension in tunisian type 2 diabetics. *La tunisie Medicale*, 88(1), 38–41. [Q4].
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62. Arfa, I., Abid, A., Nouria, S., Elloumi-Zghal, H., Malouche, D., Mannai, I., ... Zouari, B. et al. (2008). Lack of association between the angiotensin-converting enzyme gene (i/d) polymorphism and diabetic nephropathy in tunisian type 2 diabetic patients. *Journal of the Renin-Angiotensin-Aldosterone System*, 9(1), 32–36. [Q2].

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65. Arfa, I., Abid, A., Malouche, D., Alaya, N. B., Azegue, T. R., Mannai, I., . . . Blousa-Chabchoub, S. et al. (2007). Familial aggregation and excess maternal transmission of type 2 diabetes in tunisia. *Postgraduate medical journal*, 83(979), 348–351. [Q1].
66. Letac, G., Malouche, D., & Maurer, S. (2002). The real powers of the convolution of a negative binomial distribution and a bernoulli distribution. *Proceedings of the American Mathematical Society*, 130(7), 2107–2114. [Q1]. URL
67. Letac, G., & Malouche, D. (2000). The markov chain associated to a pick function. *Probability theory and related fields*, 118(4), 439–454. [Q1]. doi:10.1007/PL00008750
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